

0.1 Creating a node in the main text body

Here's a node: DEFGHI.

I used `\tikzmarknode{a}{DEFGHI}`. The name of the node is a.

Here's a node: DEFGHI.

I used `\tikzmarknode[tikzmarknode red]{a}{DEFGHI}`.

Here's a node: DEFGHI.

I used `\tikzmarknode[tikzmarknode thickred]{a}{DEFGHI}`.

Here's a node: DEFGHI.

I used `\tikzmarknode[tikzmarknode roundedthickred]{a}{DEFGHI}`.

Here's a node: **DEFGHI**.

I used `\tikzmarknode{a}{\textcolor{red}{\textbf{\Huge DEFGHI}}}`.

0.2 Create a node in the margin with boxed text

I'm going to create a tikz node in the margin using with a boxed text.

I used `\sidebox{b}{ABCDEF}`. This is a tikz node named b.

Here's how to create paragraphs.

I used `\sidebox{b}{ABCDEF \\ \\ GHIJKL}`.

TEST TEST TEST:

ABCDEF

ABCDEF

GHIJKL

I'm going to create a tikz node in the margin using with a boxed text.

ABCDEF

TEST TEST TEST: I'm going to create a tikz node in the margin

ABCDEF

using with a boxed text.

0.3 Draw arrow

Here's a node: ABCDEF.

There's a sidebox pointing to the above node. The `\DrawArrow{b}{a}` draws an arrow from node b to node a.

L^AT_EX:

```
Here's a node: \tikzmarknode{a}{ABCDEF}.\sidebox{b}{Point to ABCDEF}  
\DrawArrow{b}{a}
```

Point to ABCDEF

Example. Sidebox is lower.

ABCDEF dummy text dummy text dummy text dummy text dummy
text dummy text dummy text dummy text dummy text dummy text
dummy text dummy text dummy text dummy text dummy text dummy
text dummy text dummy text

L^AT_EX:

Point to ABCDEF

```
\tikzmarknode{a}{ABCDEF}  
dummy text dummy text dummy text dummy text dummy text dummy text  
dummy text dummy text dummy text dummy text dummy text dummy text  
dummy text dummy text dummy text dummy text dummy text dummy text  
\sidebox{b}{Point to ABCDEF}\DrawArrow{b}{a}
```

Example. Sidebox is higher.

dummy text dummy text dummy text dummy text dummy text dummy
text dummy text dummy text dummy text dummy text dummy text
dummy text dummy text dummy text dummy text dummy text dummy
text dummy text ABCDEF

Point to ABCDEF

LaTeX:

```
\tikzmarknode{a}{ABCDEF}  
dummy text dummy text dummy text dummy text dummy text dummy text  
dummy text dummy text dummy text dummy text dummy text dummy text  
dummy text dummy text dummy text dummy text dummy text dummy text  
\sidebox{b}{Point to ABCDEF}\DrawArrow{b}{a}
```

Example. Sidebox is tall.

ABCDEF

dummy text dummy text

Point to ABCDEF.
dummy text dummy
text dummy text

You can force the arrow to be horizontal using `DrawArrowH`

ABCDEF

LaTeX:

```
\tikzmarknode{a}{ABCDEF}\sidebox{b}{Point to ABCDEF. dummy text dummy text dummy text}  
\DrawArrowH{b}{a}
```

Point to ABCDEF.
dummy text dummy
text dummy text

Example. You can draw an arrow vertical-then-horizontal.

Point to ABCDEF

dummy text dummy text dummy text dummy text dummy text dummy
text dummy text dummy text dummy text dummy text dummy text
dummy text

Here's a node: ABCDEF. 

$\LaTeX$:

```
\sidebox{a}{Point to ABCDEF}

dummy text dummy text dummy text dummy text
dummy text dummy text dummy text dummy text
dummy text dummy text dummy text dummy text

Here's a node: \tikzmarknode{b}{ABCDEF}.
\DrawArrowVH{a}{b}
```

Example. You can draw an arrow horizontal-then-vertical

Point to ABCDEF

dummy text dummy text dummy text dummy text dummy text dummy
text dummy text dummy text dummy text dummy text dummy text
dummy text

Here's a node: ABCDEF.

L^AT_EX:

```
\sidebox{a}{Point to ABCDEF}

dummy text dummy text dummy text dummy text
dummy text dummy text dummy text dummy text
dummy text dummy text dummy text dummy text

Here's a node: \tikzmarknode{b}{ABCDEF}.
\DrawArrowHV{a}{b}
```


Example. You can specify displacement points for the arrow to follow using `DrawArrowPoints`. Here's an example:

Here's a node: CCCC.

DDDD

$\text{\LaTeX}$:

```
Here's a node: \tikzmarknode{cccc}{CCCC}.\sidebox{dddd}{DDDD}  
\DrawArrowPoints{dddd}{cccc}{-- ++(1,2.5) -- ++(-1.5,-1)}
```

The `--` means “line”. The `++` means “displace by”. The `-- ++(1,2.5)` means “draw a line up to a point that is (1,2.5) from the previous point”.

Instead of displacements, the points can be nodes. Here's an example:

Here's a node: CCCC.

Here's another node EEEE.

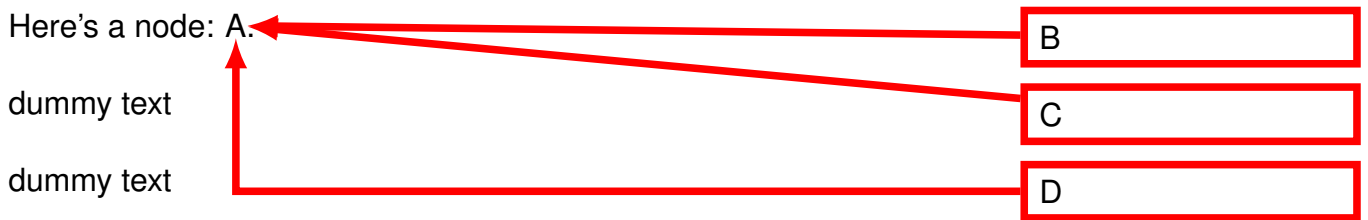
And another node FFFF.

DDDD

$\text{\LaTeX}$:

```
Here's a node: \tikzmarknode{cccc}{CCCC}.  
  
Here's another node ..... \tikzmarknode{eeee}{EEEE}.  
  
And another node \tikzmarknode{ffff}{FFFF}.\sidebox{dddd}{DDDD}  
\DrawArrowPoints{dddd}{cccc}{-- ++(1,2.5) -- (eeee) -- (ffff)}
```

Example. You can have multiple sideboxes pointing to one node.



L^AT_EX:

```
Here's a node: \tikzmarknode{a}{A}.\sidebox{b}{B}  
  
dummy text  
  
\sidebox{c}{C}  
  
dummy text  
  
\sidebox{d}{D}  
  
\DrawArrow{b}{a}  
\DrawArrow{c}{a}  
\DrawArrowHV{d}{a}
```

Example. You can have a sidebox pointing to multiple nodes.

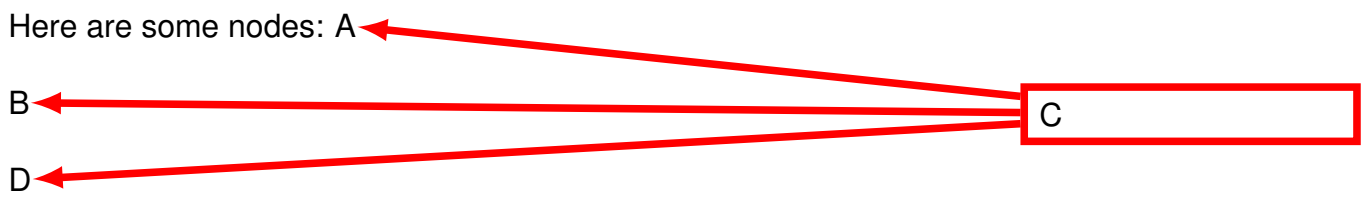
Here are some nodes: A

B

D

L^AT_EX:

```
Here are some nodes:  
\tikzmarknode{a}{A}  
  
\tikzmarknode{b}{B}\sidebox{c}{B}  
  
\tikzmarknode{c}{C}.  
\DrawArrow{c}{b}  
\DrawArrow{c}{d}
```



C

0.4 Tikz node inside code

Here's an example where the tikz node is inside code:

CCC

```
#include <iostream>

int main()
{
    return 0; // a is a node containing return
}
```

$\text{\LaTeX}$:

```
\sidebox{b}{CCC}
\begin{consolethre}[escapeinside=||]
#include <iostream>

int main()
{
    |\tikzmarknode{a}{return}| 0; // a is a node containing return
}
\end{consolethre}
\DrawArrow{b}{a}
\end{consolethre}
\DrawArrow[red]{start12}{end12}
```

NOTE: console and Verbatim cannot be used. consolethre uses the listings package. The escapeinside|| basically means whatever is within |...| is treated as latex command. If the C++ or python code contains |, use another pair of characters.

Example: Here's an example with tikz node inside code and the node contains a backslash character:

```
#include <iostream>

int main()
{
    std::cout << "hello world\n";
    return 0;
}
```

Point to a string

L^AT_EX:

```
\sidebox{a}{Point to a string}
\begin{consolethreethree}[escapeinside=|]
#include <iostream>

int main()
{
    std::cout << |\tikzmarknode[tikzmarknode thickred]{b}{"hello world\char'\n"}|;
    return 0;
}
\end{consolethreethree}
\DrawArrow{a}{b}
```

Note the proper way of inserting the backslash character for the new-line character.

To make it easier, I have a macro for the backslash using typewriter font: `\bstt` (memory aid: `bstt` = backslash typewriter text).

Example. `ABC\DEF` is done using `ABC\bstt DEF`.

Example: Here's the earlier example using the `\bstt` macro:

Point to a string

```
#include <iostream>

int main()
{
    std::cout << "hello world\n";
    return 0;
}
```



L^AT_EX:

```
\sidebox{a}{Point to a string}
\begin{consolethre}[escapeinside=|]
#include <iostream>

int main()
{
    std::cout << |\tikzmarknode[tikzmarknode thickred]{b}{hello world\bstt n}|;
    return 0;
}
\end{consolethre}
\DrawArrow[{a}]{b}
```

Example: You might want to align a sidebox with a specific line of code. The best is to put the sidebox next to that line (and maybe put the draw arrow there as well).

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "hello world\n";
```

```
    return 0;
```

```
}
```

Point to a string

L^AT_EX:

```
\begin{consolethre}[escapeinside=|]|
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << \tikzmarknode[tikzmarknode thickred]{b}{"hello world\bstt n"};|\sidebox{a}{Point to a string}
```

```
    return 0;
```

```
}
```

```
\end{consolethre}
```

```
\DrawArrow{a}{b}
```

It might not be completely aligned. You can then shift the sidebox vertically – see next section.

Also, note that since the sidebox is drawn inside `consolethre`, the font used is typewriter font. In the sidebox, you switch to normal font using `\textnormal`.

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "hello world\n";
```

```
    return 0;
```

```
}
```

Point to a string

L^AT_EX:

```
\begin{consolethre}[escapeinside=|]|
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << \tikzmarknode[tikzmarknode thickred]{b}{"hello world\bstt n"}\sidebox{a}{\textnormal{Point to a string}}|;
```

```
    return 0;
```

```
}
```

```
\end{consolethre}
```

```
\DrawArrow{a}{b}
```

0.5 Shift sidebox vertically

You can move the sidebox vertically. Here's an example where the sidebox is not shifted:

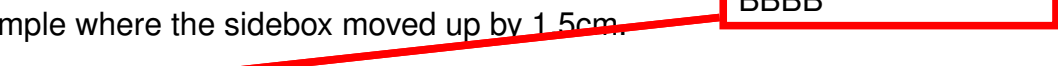
Here's a node AAAA.



BBBB

dummy text dummy text dummy text dummy text dummy text dummy text
text dummy text dummy text dummy text dummy text dummy text
dummy text

Here's the same example where the sidebox moved up by 1.5cm.



BBBB

Here's a node AAAA.

$\text{\LaTeX}$:

```
Here's a node \tikzmarknode{b}{AAAA}\sidebox[-1.5cm]{a}{BBBB}.  
\DrawArrow{a}{b}
```

Note the `[-1.5cm]`.

Example:

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "hello world\n";
```

```
    return 0;
```

```
}
```

Point to a string

L^AT_EX:

```
\begin{consolethree}[escapeinside=||]
```

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << |\tikzmarknode[tikzmarknode thickred]{b}{\"hello world\\bstt n\"}\\sidebox[-0.5cm]{a}{\\textnormal{Point to a string}}|;
```

```
    return 0;
```

```
}
```

```
\end{consolethree}
```

```
\DrawArrow{a}{b}
```

0.6 Reuse tikz node names

You can reuse tikz node names.

0.7 Warning on disappearing tikz diagrams

Watch out: if a tikz diagram is too small and too close to the bottom of a page, the diagram might be accidentally truncated/removed by latex.

0.8 Minor variations on draw arrow



$\LaTeX$:

```
\tikzmarknode{b}{B}\sidebox{a}{B}\DrawArrow[dashed]{a}{b}
```



$\LaTeX$:

```
\tikzmarknode{b}{B}\sidebox{a}{A}\DrawArrow[dashed, red]{a}{b}
```



$\LaTeX$:

```
\tikzmarknode{b}{A}\sidebox{a}{A}\DrawArrow[blue]{a}{b}
```

0.9 xsidebox

The `\xsidebox` macro allows you to specify x and y shifts and some node attributes.

Sometimes it's difficult/impossible to shift the node. There's some kind of constraint on the position.

Example: Some text. Some text. Some text.

`\xsidebox{a}{A}` is the same as `\sidebox{a}{A}`

A

B

C

D

$\LaTeX$

```
\tikzmarknode{a}{A}\xsidebox{b}{B}
\tikzmarknode{c}{C}\xsidebox{d}{D}
```

Example: Some text. Some text. Some text.

`\xsidebox{a}{A}[1cm]` is the same as `\sidebox{a}{A}` but shifted by 1cm on *x*-axis.

A



C

$\LaTeX$

```
\tikzmarknode{a}{A}\xsidebox{b}{B}

\tikzmarknode{c}{C}\xsidebox[-1cm]{d}{D}
```

Example: Some text. Some text. Some text.

`\xsidebox[1cm][2cm]{a}{A}` is the same as `sidebox{a}{A}` but shifted by 1cm on x -axis and 2cm on the y -axis.

A



C

$\LaTeX$

```
\tikzmarknode{a}{A}\xsidebox{b}{B}
\tikzmarknode{c}{C}\xsidebox[-1cm][1cm]{d}{D}
```


Example: Some text. Some text. Some text.

`\xsidebox[1cm][2cm]{a}{A}` is the same as `sidebox{a}{A}` but shifted by 1cm on *x*-axis and 2cm on the *y*-axis.

A

D

B

C

Example: Some text. Some text. Some text.

`\xsidebox[1cm][2cm][blue]{a}{A}` is the same as `sidebox{a}{A}` but shifted by 1cm on x -axis and 2cm on the y -axis and add blue to node attributes.

A



C

$\LaTeX$

```
\tikzmarknode{a}{A}\xsidebox{b}{B}
\tikzmarknode{c}{C}\xsidebox[-1cm][1cm][blue]{d}{D}
```